

Dominic Bair

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Education

- 2022 – ... 📖 **Ph.D. candidate, Mathematics**, University of Tennessee Knoxville (UTK).
- 2020 – 2022 📖 **M.S. Mathematics**, Montana State University (MSU).
- 2016 – 2020 📖 **B.S. Mathematics; Minor: Physics**, Montana State University.

Publications

- 1 D. Bair, S. Jana, and Y. Qing, “First-passage percolation, non-positive curvature, and radial maps,” *Preprint*, 2025. 🔗 URL: <https://arxiv.org/abs/2512.05720>.
- 2 D. Bair, “DVM: A deep learning algorithm for minimizing functionals,” *Theses and Dissertations at Montana State University (MSU)*, 2022. 🔗 URL: <https://scholarworks.montana.edu/xmlui/handle/1/16840>.
- 3 C. Potts, J. Schearer, T. A. Sebrell, *et al.*, “MNPmApp: An image analysis tool to quantify mononuclear phagocyte distribution in mucosal tissues,” *Cytometry Part A*, vol. 101, no. 12, pp. 1012–1026, 2022. 🔗 URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/cyto.a.24657>.

Employment History

Paid Positions

- 2022 – ... 📖 **Graduate Teaching Assistant**, UTK Department of Mathematics.
- 2023 – 2024 📖 **Climbing Wall Attendant/Setter**, UTK RECSports.
- 2020 – 2022 📖 **Graduate Teaching Assistant**, MSU Department of Mathematical Sciences.
- 2015 – 2022 📖 **Laborer**, 3 Brothers Plumbing and Heating.
- 2020 📖 **Laborer**, Peak Property Management.
- 2019 – 2020 📖 **Undergraduate Research Assistant**, MSU Department of Mathematical Sciences.
- 2018 – 2019 📖 **Undergraduate Research Assistant**, MSU Department of Physics.

Volunteer Positions

- 2026 – ... 📖 **Senator**, UTK Graduate Student Senate.
- 📖 **Leader**, UTK Math Graduate Student Seminar.
- 2018 – 2020 📖 **Assistant**, National Center for Unwanted Firearms.
- 2015 – 2017 📖 **Firefighter**, Clancy Volunteer Fire Department.

Courses Taught

- M115 📖 **Statistical Reasoning**, UTK.
- M125 📖 **Basic Calculus**, UTK.
- M113 📖 **Mathematical Reasoning**, UTK.
- M119 📖 **College Algebra**, UTK.
- M171 📖 **Calculus I**, MSU.
- M151 📖 **Precalculus**, MSU.
- M161 📖 **Survey of Calculus**, MSU.

Employment History (continued)

M121  College Algebra, MSU.

Research Interests

My research interests lie at the intersection of geometric group theory, probability, dynamical systems, and applied stochastic modeling. I am currently studying how finite-state automata and Markov measures describe geodesics and boundary measures in relatively hyperbolic groups, with particular interest in Patterson–Sullivan theory. I am also interested in random metric geometry, including how first-passage percolation changes hyperbolicity, Morse geodesics, and other manifestations of nonpositive curvature. Alongside these theoretical questions, I use stochastic calculus, numerical simulation, and statistical calibration to investigate derivative pricing and hedging, including quanto options. Broadly, I seek structural principles connecting geometry, randomness, dynamics, and computation across pure and applied mathematics.

Skills

Coding/Software  Codex, MATLAB, Python, $\text{\LaTeX}$, Julia, Windows, MacOS, Linux.
Misc.  Technical problem solving, leadership, academic research, teaching, first aid.

Miscellaneous Experience

Awards and Achievements

2026  Elected as UTK Graduate Student Senator.
  UTK Graduate Student Academic Achievement Award.
2025  UTK Oral Specialty Exam, passed.
  UTK Preliminary Exam in Topology, passed.
  UTK Preliminary Exam in Stochastics, passed.
2020  NSF GRFP Reviews, Excellent, Excellent, Very Good. (Fellowship not received).
  NSF and AWM Travel Grant, to present at Data Science and Image Analysis Conference of the Pacific Northwest.
2019 – 2020  INBRE Grant, to support the Spatial Statistics in Histology Images research. NIH funded, MSU based grant
2016 – 2020  MSU Premier Scholarship, for demonstration of academic promise, leadership experience and other honors.
2019  Superior Achievement in the Field of Mathematics, recognition by the National Honorary Mathematics Society, *IIME*, for superior achievement in the field of mathematics.
2016  Eagle Scout, the highest achievement that can be obtained in the Scouts of America.

Talks Given

2026  Ordered frequency in relatively hyperbolic groups, UTK Topology Seminar.
  Quanto Options, Erdős Institute Quant Finance Bootcamp.
  First-passage percolation, non-positive curvature, and radial maps, YGGT XIV.
  Graphons and random graphs, UTK Graduate Student Seminar.
2025  Introduction to first-passage percolation and subadditive processes, UTK Probability Seminar.
  First-passage percolation on hyperbolic graphs: bi-infinite geodesics, radial maps, and infinitely many thick triangles, UTK Topology Seminar.

Miscellaneous Experience (continued)

- 2022  **First-passage percolation on a hyperbolic graph admits bi-infinite geodesics**, UTK Oral Specialty Exam.
- 2022  **Deep variational methods**, MSU Master's Thesis Defense.
- 2021  **Deep variational methods**, MSU Poster Slam.
-  **Deep variational methods**, Walk the Walk Chalk the Chalk.
- 2020  **Spatial statistics in histology images**, SIAM Northern States Section Student Chapters Conference.
-  **Spatial statistics in histology images**, Data Science and Image Analysis Conference of the Pacific Northwest.
- 2019  **Spatial statistics in histology images**, MSU Winter Research Celebration.

Reading Courses

- 2026  **Big Mapping Class Groups**, UTK.
-  **Fundamentals of Stein's Method (Ross)**, UTK.
- 2023  **Monte Carlo Strategies in Scientific Computing (Liu)**, UTK.

Certifications

- 2026  **Quant Finance Bootcamp**, The Erdős Institute.
- 2023  **CPR and First Aid**, American Red Cross.